Mansoor Ahmed

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Education

Georgia State University / Georgia Tech (Joint), Atlanta, GA 2024 – 2028 (expected)

- PhD Candidate in Computer Science/Machine Learning (+ M.Sc CS). GPA: 4.0/4.0.

University of Engineering and Technology, Lahore Dec 2017 – Aug 2021

- B.Sc. Electrical Engineering. Full scholarship recipient by Higher Education Commission Pakistan

Research Experience

Algorithmic Biology Lab, Georgia State University — Graduate Research Assistant Aug 2024 – present

- Advised by **Prof. Murray Patterson** & **Prof. Hemanth Venkateswara**.
- Developing theoretical generative modeling frameworks with applications to biomolecule/ antibody design.
- First-author papers published at TMLR, and ICLR, ICML, and NeurIPS workshops (see below).

Lab for Understanding Network Effects (LUNE), UCLA — Research Fellow Aug 2025 – May 2026

- Advised by **Prof. Jennifer Wilson**. Drug combination modeling for cancer subtypes via network methods.

Data Analytics Lab, LUMS — Research Assistant (Lahore, Pakistan) Dec 2022 – June 2024

- Advised by **Prof. Imdadullah Khan**. Graph-based methods for antigen-antibody interactions.

Smart Data Systems Lab, LUMS — Research Assistant (Lahore, Pakistan) Nov 2021 – Nov 2022

- Advised by **Prof. Zubair Khalid**. Software development and database management for IoT devices.

Publications

Accepted Publications

1. **ChiMERa-Bench: A Benchmark Dataset for Epitope-specific Antibody Design**
   M. Ahmed, N. Taj, I. U. Khan, H. Venkateswara, M. Patterson.
   Transactions on ML Research (TMLR, 2026); 42+ GitHub stars [arXiv, OpenReview]

2. **Latent-Kernel Discrete Diffusion for Few-Step Generation**
   M. Ahmed, Y.T. Fan, H. Venkateswara, M. Patterson. *NeurIPS'26 BeNTo Workshop.* [arXiv, OpenReview]

3. **EvoStruct: Bridging Evolutionary and Structural Priors for Antibody CDR Design via Protein Language Model Adaptation**
   M. Ahmed, S. Lee, U. Khayaz, M. Patterson. *ICML'26 GenBio Workshop.* [arXiv, OpenReview]

4. **ConTact: Contact-First Antibody CDR Design via Explicit Interface Reasoning**
   M. Ahmed, S. VonBank, Nadeem Taj, S. Lee, N. Jan, **M. Patterson.** *ICML'26 GenBio Workshop.* [arXiv, OpenReview]

5. **Improved Graph-based Antibody-aware Epitope Prediction with PLM Embeddings**
   M. Ahmed, S. Ali, A. Jan, I. U. Khan, M. Patterson. *ICCABS'25. Springer LNBI post-proceedings.* [Springer, bioRxiv]

6. **Robust Brain Age Estimation via Regression Models and MRI-derived Features**
   M. Ahmed, U. Sardar, S. Ali, S. Alam, M. Patterson, I. U. Khan. *ICCCI'23. Springer CCIS.* [arXiv, Springer]

Preprints & Under Review

1. **MLD3: Mixture-of-Latents for Few-Step Discrete Diffusion Distillation**
   M. Ahmed, S. Tang, H. Venkateswara, M. Patterson. *Under review at ICLR'27.* [OpenReview]

2. **Domino: Joint Unmasking via First-order Markov Chain for Few-Step Masked Diffusion**
   M. Ahmed, L. Chebbah, J. Dinh, H. Venkateswara, M. Patterson. *Under review at TMLR, 2026.* [OpenReview]

3. **EpiFormer: Learning Antigen-Antibody Interactions for Epitope Prediction**
   M. Ahmed, H. Chai, H. Wang, H. Venkateswara, M. Patterson. *Under review at ICLR'27.* [arXiv, OpenReview]

4. **AgForce Enables Antigen-conditioned Generative Antibody Design**
   M. Ahmed, M. Patterson. *Under review at AAAI'27 (phase 2).* [arXiv]

Teaching Experience

Advanced Machine Learning (CSC 8850) — Georgia State University — Spring 2025; Fall 2025, 2026

- Graduate Teaching Assistant with **Prof. Sergey Plis**. Held office hours; created and graded problem sets and exams; delivered lectures on manifold learning.

Computer Org. & Programming (CS 3210) — Georgia State University — Summer 2025, Spring 2026

- Teaching Assistant. Assembly language and computer architecture.

Spatial and Scientific Databases (CSC 8713) — Georgia State University — Fall 2024

Mathematical Foundations for Machine Learning (EE 212) — LUMS — Fall 2022

Honors & Awards

Molecular Basis for Disease (MBD) Fellowship, Georgia State University — 2025 – present

- Fellowship awarded to GSU PhD students for novel and wet-lab collaborative work on Molecular design

Warren Alpert Computational Biology and AI Network Fellow, UCLA — 2025 – 2026

Computational Genomics Summer Institute (CGSI), UCLA — 2025

EMBO Latin American Workshop on AI for Protein Design, Chile — 2025

International Summer School in Biomedical Engineering, Max Planck Institute — 2023

Higher Education Commission Undergraduate Scholarship, Pakistan — 2017 – 2021

Reviewing & Service

Workshops: GenBio @ ICML'26, GEM @ NeurIPS'26, ML4H @ NeurIPS'26, BeNTo @ NeurIPS'26

Conferences: AAAI'27, ICLR'27, ISBRA'25;26, ICCABS'25

Journals: PeerJ Computer Science, TMLR

Skills

ML/DL: PyTorch, PyTorch Geometric, e3nn, PyTorch Lightning, Hugging Face, WandB

Protein ML: ESM-2, AntiBERTy, ProteinMPNN, AlphaFold2, OpenFold, BoltzGen, RFdiffusion

Structural Biology: BioPython, PyMOL, RosettaDock, AutoDock, BLAST, ANARCI

Programming: Python, C++, CUDA, SQL, Git, Unix/Shell, LaTeX, SLURM, Docker